SWEEP rule on the KD-Toric code

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1 The Turns Idea.

The main problem we have is that, for now, the history is defined only for configurations that, at time t, see a non-trivial syndrome. What we can do is imagine that at each turn we flip the bits according to a fixed order. We think of the basic time unit as a turn, and each real-time iteration contains tn^d turns. Now, if a bulk of bits A is flipped at time t, then there must be a turn $\tau \leq tn^d$ such that at that turn there was a configuration that has a history.

Now we would like to describe what the history of that configuration looks like, and we do that by connecting the turn τ to the closest $t'n^d$. The idea is that we cann't ensure the shrinking lemma. Yet we can tell for sure that if there is expansion then its limited.

I have also the felling that the shrinking works, and the idea goes as follow, assume that $v \in \Gamma^{(t-1)}$ gets the maximum of L_i either we stop before flipping a bit that is 'supported' on v or not, in the first case, we get that any u that is added by v' might has $L_i(u) = L_i(v') < L_i(v)$, in the second case we have already handled.

The turn machinery makes sense only in the finite case. Yet, we can back to the Tree Separator Section, and show that with probability

$$n^d \cdot tn^d q^n$$

We don't have a big trees which implies independence. Then we can show that with probability:

$$n^d t(n^d)^2 q'^{\delta n}$$

There was no turn in which it had a configuration that was connected at the time, saw a syndrome, and had a side-length that behaved like $\Theta(n)$.

[COMMENT] The model has to redefined again such it will be clear that outcomes are possibles running of the system. Currently this can not be inferred.

Claim 1.1. Let $\omega \in \Omega$ be an outcome,